International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

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Registration Number - C11/26/PCDFIR/17022

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Course Code - CIP-B101

Course Title - Basic Computer Skills for Digital Forensics

Tools Used – Command Prompt, Power Shell, WinHex

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

Student Practical Laboratory Manual

Course Code	CIP-B101
Course Title	Basic Computer Skills for Digital Forensics
Lab Code	Lab 1B
Lab Title	PC System Fundamentals, Disk Sectors and Partition Analysis
Estimated Time	3-4 hours
Mode	Individual practical lab with screenshots and written calculations
Required Output	A structured lab report submitted in PDF format
Source Material	Computer System Introduction slide deck

Part A - System Identification and Evidence Layer Mapping

In this section, you will identify the major computer system layers and explain how each layer can produce forensic evidence.

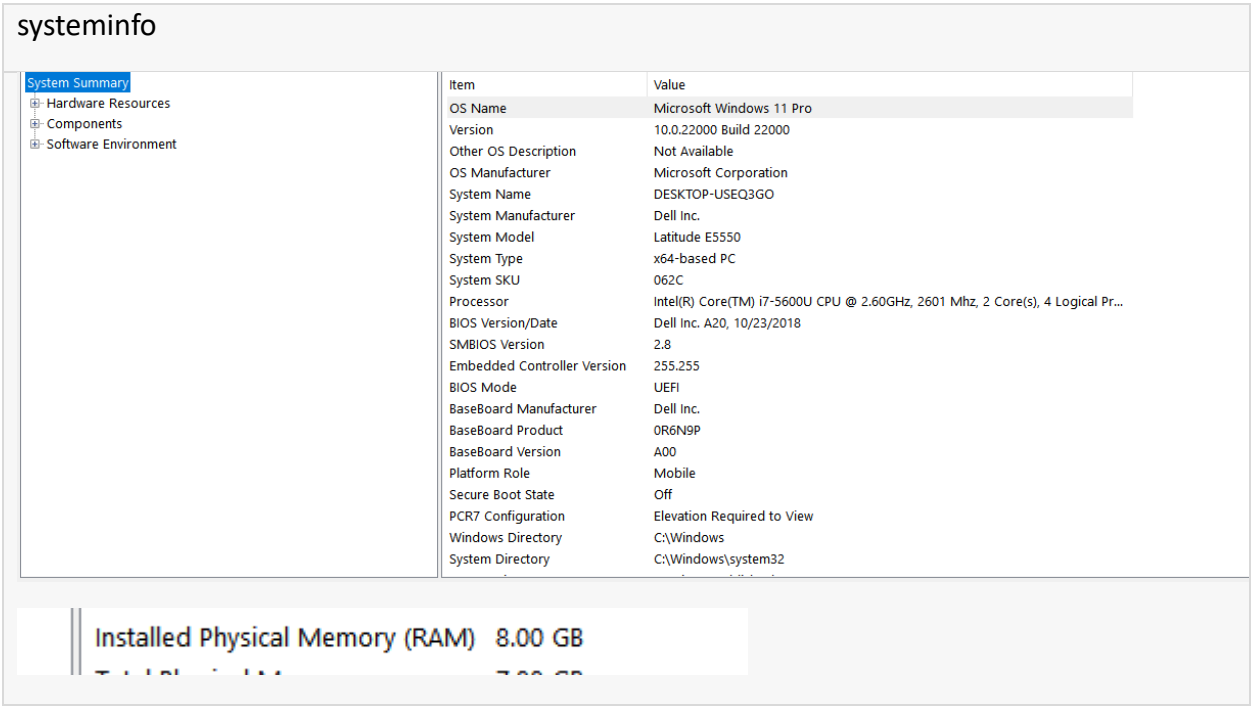
A1. Capture basic system specification

On Windows, open System Information by pressing Windows + R, typing msinfo32, and pressing Enter.

Capture a screenshot showing OS Name, Version, System Type, Processor, BIOS Mode, Installed Physical Memory, and Secure Boot State if available.

Also run the command below in Command Prompt and capture the first page of output.

```
systeminfo
```



```
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ASHLEY>systeminfo

Host Name:                 DESKTOP-USEQ3GO
OS Name:                   Microsoft Windows 11 Pro
OS Version:                10.0.22000 N/A Build 22000
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          ASHLEY
Registered Organization:
Product ID:                00331-10000-00001-AA508
Original Install Date:     8/24/2023, 11:28:27 PM
System Boot Time:          8/19/2026, 10:20:30 PM
System Manufacturer:       Dell Inc.
System Model:              Latitude E5550
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 61 Stepping 4 GenuineIntel ~500 Mhz
BIOS Version:              Dell Inc. A20, 10/23/2018
Windows Directory:         C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+01:00) West Central Africa
Total Physical Memory:     8,072 MB
Available Physical Memory: 2,018 MB
Virtual Memory: Max Size:  12,238 MB
Virtual Memory: Available: 3,186 MB
Virtual Memory: In Use:    9,052 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\DESKTOP-USEQ3GO
Hotfix(s):                 6 Hotfix(s) Installed.
                           [01]: KB5030650
                           [02]: KB5030842
                           [03]: KB5008295
                           [04]: KB5011048
                           [05]: KB5031358
                           [06]: KB5031591
```

A2. Capture storage information

Open Command Prompt or PowerShell.

Run one of the command sets below. Capture screenshots of storage model, size, partitions, and file systems.

If a command is not supported on your system, use the alternative command.

```
wmic diskdrive get model,size,serialnumber
wmic logicaldisk get caption,filesystem,size,freespace,volumename

PowerShell alternative:
Get-Disk
```

```
Get-Partition
Get-Volume
```

```
C:\Users\ASHLEY>wmic diskdrive get model,size,serialnumber
Model                SerialNumber  Size
TOSHIBA MQ01ABF050  53FEW8KCT     500105249280
```

A3. Map evidence sources to system layers

Create a table in your report showing at least five system layers/sources and the type of evidence that can be generated from each one.

Use this guide: User activities = Applications + Operating System + Disk/Memory.

Explain why evidence may exist in more than one layer.

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Opened files, searched terms, typed usernames	Recent files, browser history, shell history	Links user action to system activity
Application layer	Office document metadata, browser cache, chat logs	Application folders and databases	Shows what software created or used evidence
OS layer	Event logs, installed programs, user profiles	Windows logs, registry, Linux /var/log	Shows system events and account activity
Disk layer	Deleted files, file fragments, partitions	File system and unallocated space	Supports recovery and timeline analysis
Memory/process layer	Running malware, PIDs, network connections	RAM capture, process list	Shows live system behavior

A users actions affects different layers which leaves traces or evidence of the actions performed, e.g. if a user opens a document, the user layer shows it in the recent files list. The application

layer shows it in the app's metadata or cache. The OS layer logs the process launch and file access in event logs or the registry. The disk layer holds the file and its timestamps in the file system. The memory layer may hold the running process and the document's contents while it is open. This overlap matters because evidence can be deleted or altered at one layer but survive at another, and matching evidence across layers lets an investigator corroborate a timeline and detect tampering.

Part B - Program, Process and Thread Investigation

This section links the slide concept of program, process, and thread to forensic observation.

B1. Observe running processes

Open Task Manager on Windows, or use Command Prompt/PowerShell.

Capture a screenshot showing running processes.

Select three processes and record their process names, PID values, and likely purpose.

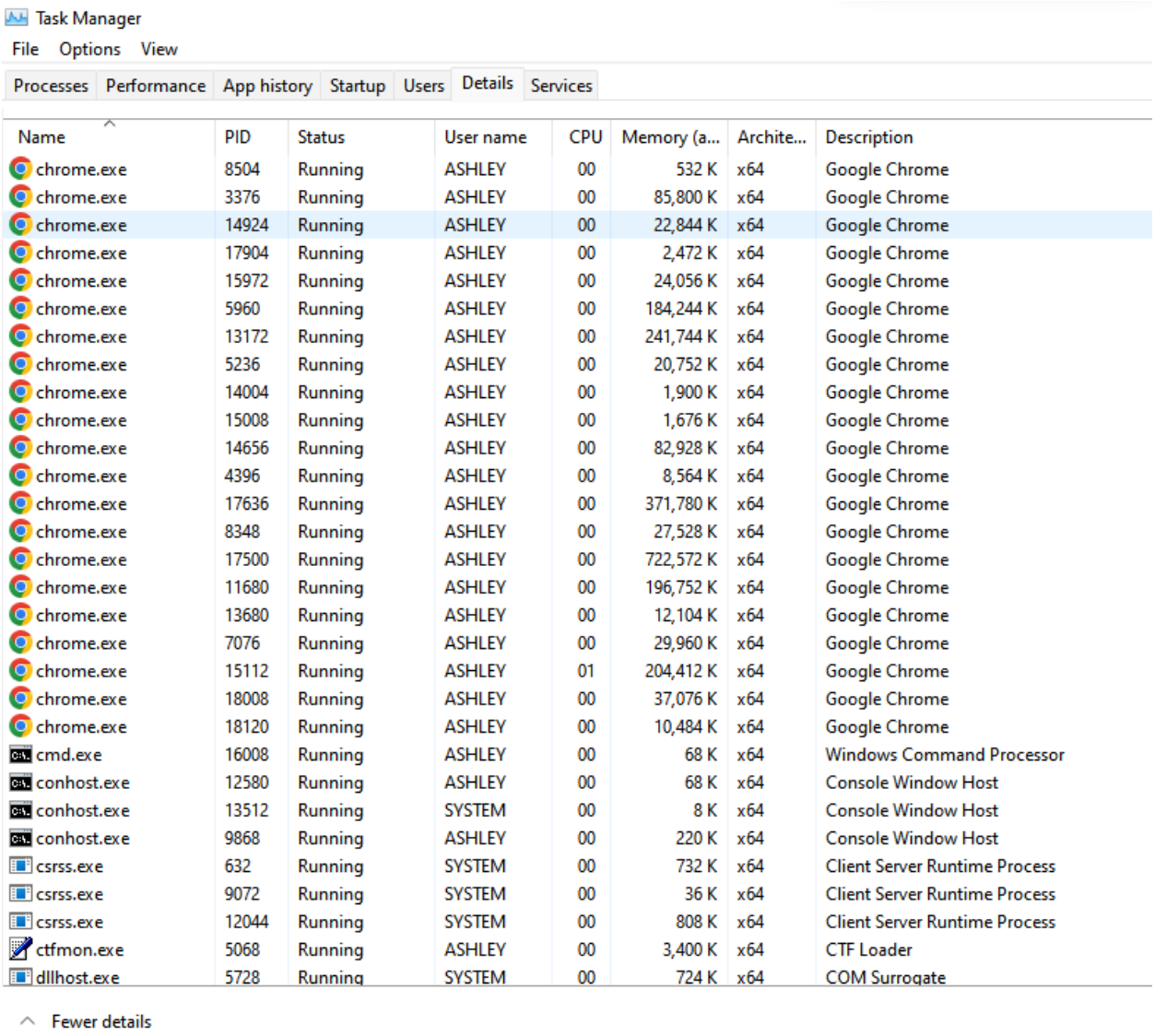
```
tasklist

PowerShell alternative:
Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet
```

B2. Explain the difference

In your report, explain the difference between a program, a process, and a thread in your own words.

Explain why process information may be useful during malware analysis or live forensic triage.



Process	PID	Likely purpose
conhost.exe	9868	The Console Window Host
chrome.exe	18120	Web browser
cmd.exe	16008	The Windows Command Prompt

Program, process and thread

- Program: A program is a passive file stored on disk, such as notepad.exe. It contains instructions but isn't doing anything.
- Process: A process is a program that has been loaded into memory and is running. It has its own PID, memory space, and resources like open files and network connections. You can run the same program as several separate processes.

- Thread: A thread is the smallest unit of execution inside a process. A process can have many threads, which share its memory and resources but run independently. A browser, for example, may use separate threads for rendering, networking, and downloads.

Process information matters in malware analysis and live triage running because processes show what is executing immediately, including malware that never touches the disk. Process names, PIDs, parent-child relationships, CPU use, and network connections can expose suspicious activity.it is volatile evidence and very useful for investigations.

Part C - Disk Sector and Disk Size Calculation

Key formula
Disk or partition size = number of sectors x bytes per sector. To convert bytes to GB, divide by 1024 x 1024 x 1024.

C1. Reproduce the slide calculation

Use the values from the slide to calculate the total disk size: sectors per track = 400, heads = 12, cylinders = 17000, and bytes per sector = 512.

Item	Formula / Value	Answer
Total sectors	400 x 12 x 17000	81,600,000 sectors
Total bytes	81,600,000 x 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 / 1024	40,800,000 KB
Megabytes	40,800,000 / 1024	39,843.75 MB
Gigabytes	39,843.75 / 1024	38.91 GB

C2. Calculate your own disk or partition size

Use your partition table or disk details to identify at least two partitions.

For each partition, record the start sector, number of sectors, sector size, and file system where available.

Calculate the size of each partition in GB using the formula above.

Your calculation must be shown step-by-step. Screenshots alone are not enough.

Example calculation:
Partition size in GB = sector_count x 512 / 1024 / 1024 / 1024

If sector_count = 83,881,984:
83,881,984 x 512 / 1024 / 1024 / 1024 = 39.99 GB

Partition	File system	Start sector	Number of Sectors	Sector size
1	FAT32	2,048	204,800	512
3	NTFS	239,616	739,796,992	512
4	Ext4	740,036,608	235,520,000	512
5	NTFS	975,556,608	1,214,464	512

Partition 1

Number of sectors = 206,848 - 2,048 = 204,800 sectors.

Size in bytes = 204,800 × 512 = 104,857,600 bytes.

KB = 102,400 KB

MB = 100 MB

Size in GB = 104,857,600 ÷ 1,073,741,824.

Partition 1 = 0.0977 GB.

Partition 3

Number of sectors = 740,036,608 - 239,616 = 739,796,992.

Size in bytes = 739,796,992 x 512 = 378,776,059,904 bytes

KB = 378,776,059,904 / 1024 = 369,898,496 KB

MB = 369,898,496 / 1024 = 361,229 MB

Size in GB = 361,229 / 1024 = 352.76 GB

Partition 3 = 352.76 GB.

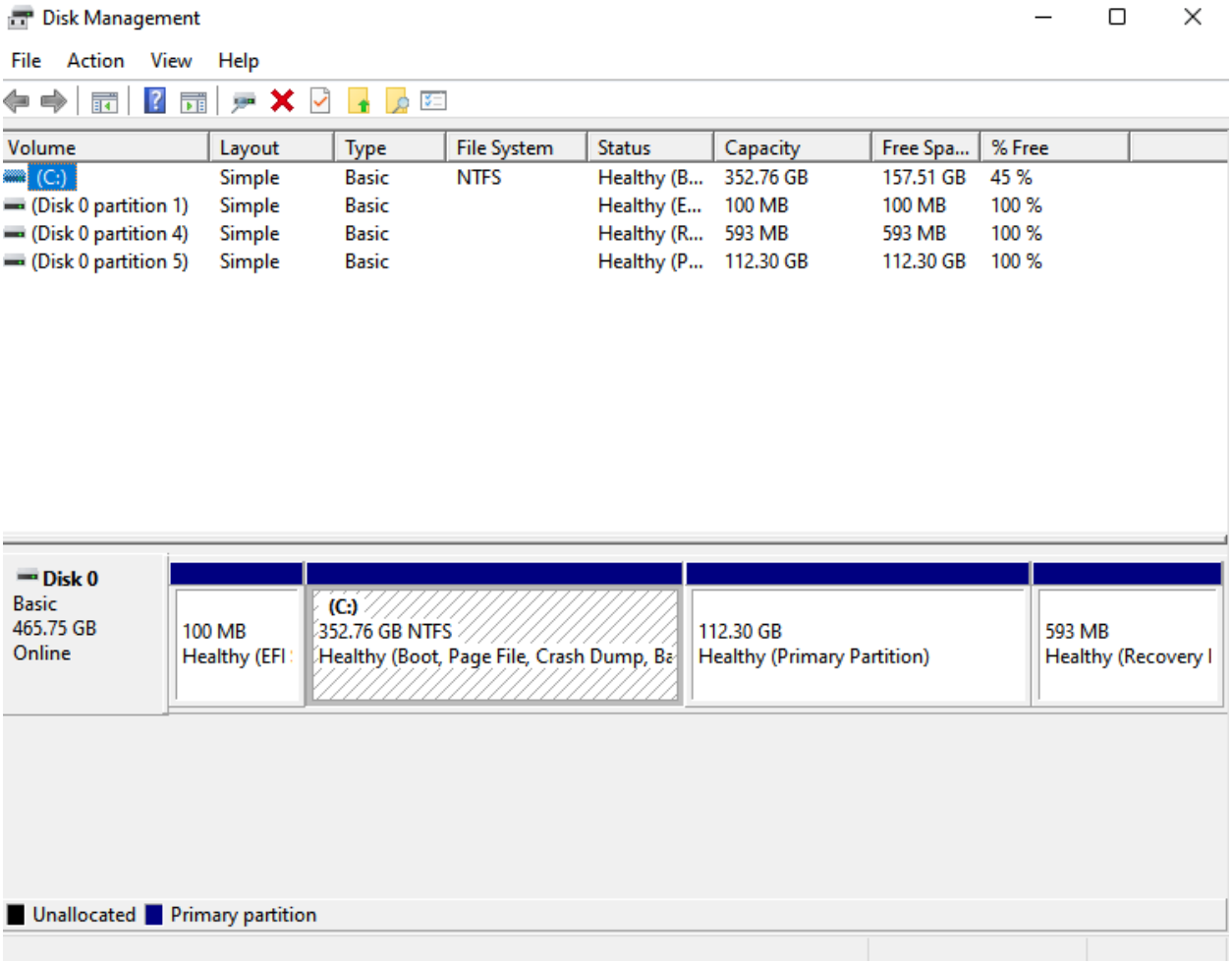
Part D - Safe Partition Table Inspection

Choose either Windows or Linux. You may do both for extra practice. You must not modify the disk.

D1. Windows method - Disk Management, PowerShell and DiskPart

Open Disk Management by pressing Windows + R, typing diskmgmt.msc, and pressing Enter.

Capture a screenshot showing your disk layout and visible partitions.



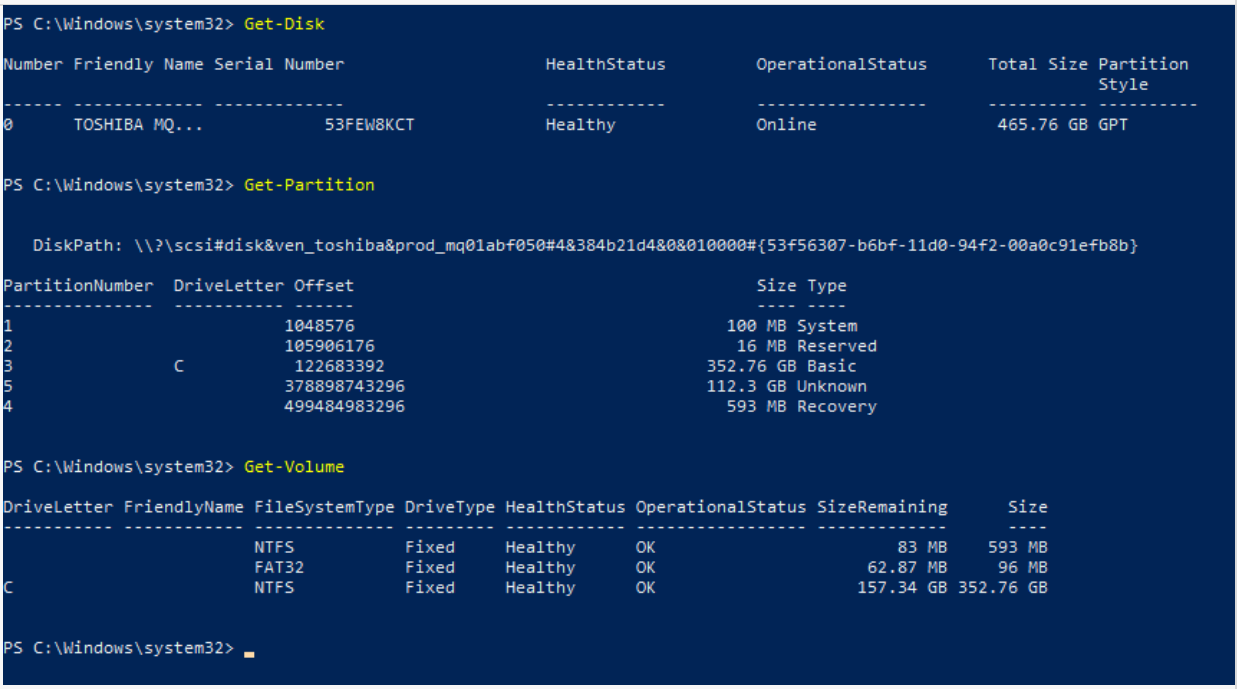
Open PowerShell as normal user or administrator if required and run the safe read-only commands below.

Open Command Prompt as administrator and run DiskPart only for listing disks and partitions. Do not run create, delete, clean, format, active, or convert commands.

```
PowerShell:
Get-Disk
Get-Partition
Get-Volume

DiskPart safe listing only:
diskpart
list disk
```

select disk 0
list partition
exit



```
C:\Windows\system32>diskpart

Microsoft DiskPart version 10.0.22000.653

Copyright (C) Microsoft Corporation.
On computer: DESKTOP-USEQ3GO

DISKPART> list disk

  Disk ###  Status         Size     Free     Dyn  Gpt
  --------  -------------  -------  -------  ---  ---
  Disk 0    Online          465 GB  1024 KB        *

DISKPART> select disk 0

Disk 0 is now the selected disk.

DISKPART> list partition

  Partition ###  Type              Size     Offset
  -------------  ----------------  -------  -------
  Partition 1    System             100 MB  1024 KB
  Partition 2    Reserved            16 MB   101 MB
  Partition 3    Primary            352 GB   117 MB
  Partition 5    Unknown            112 GB   352 GB
  Partition 4    Recovery           593 MB   465 GB

DISKPART> exit

Leaving DiskPart...

C:\Windows\system32>
```

D2. Linux/Kali method - read-only inspection

Open Terminal in your Linux/Kali VM or host.

Run the commands below and capture screenshots showing disk, partition, size, and file system information.

Do not run mkfs, fdisk write commands, parted write commands, dd write commands, or mount read-write commands for this lab.

```
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
sudo fdisk -l
sudo parted -l
```

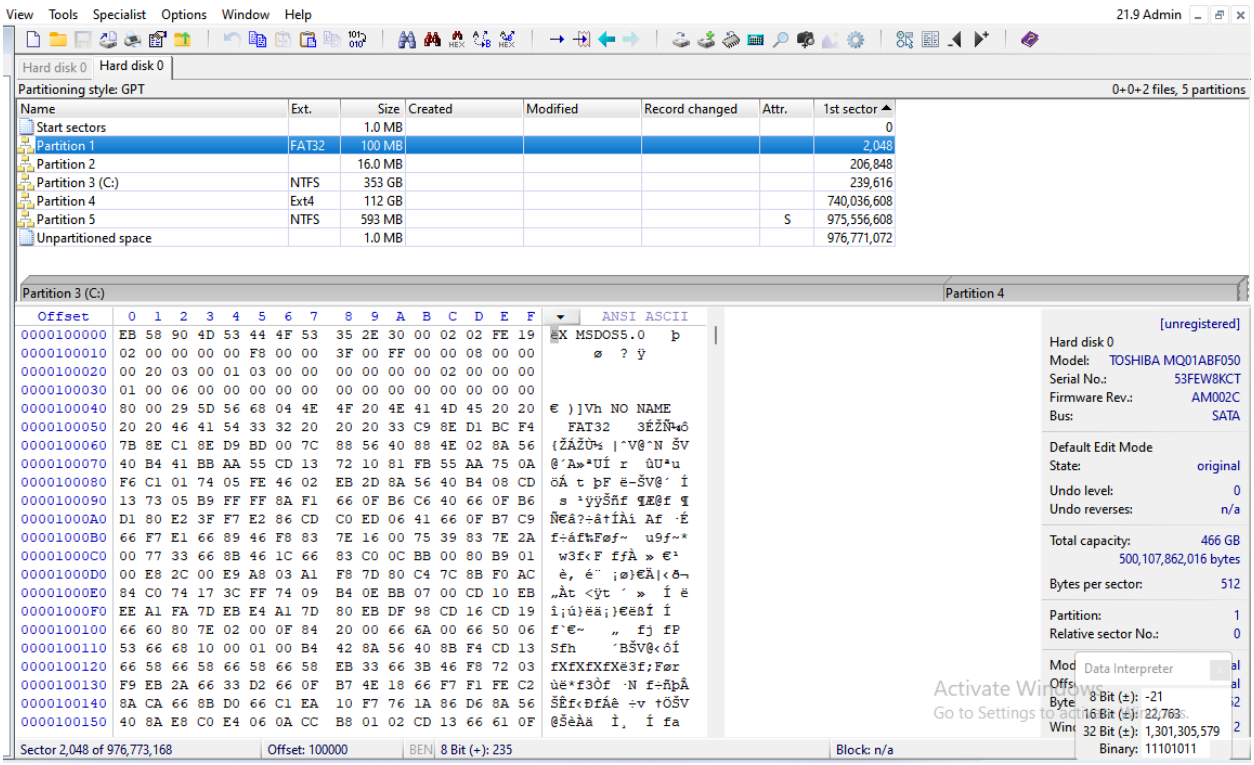
```
blkid
df -h
```

D3. Optional WinHex or forensic viewer method

If your instructor permits, open the physical disk or disk image in a forensic/hex viewer in read-only mode.

Navigate to sector 0 and capture the first sector where the MBR/protective MBR may be visible.

Identify whether the disk appears to use MBR or GPT. If you cannot safely view this, skip the optional step.



It is GPT

Part E - MBR, GPT and Boot Process Explanation

E1. Explain the boot process

Create a simple flow diagram or numbered list explaining how a PC boots.

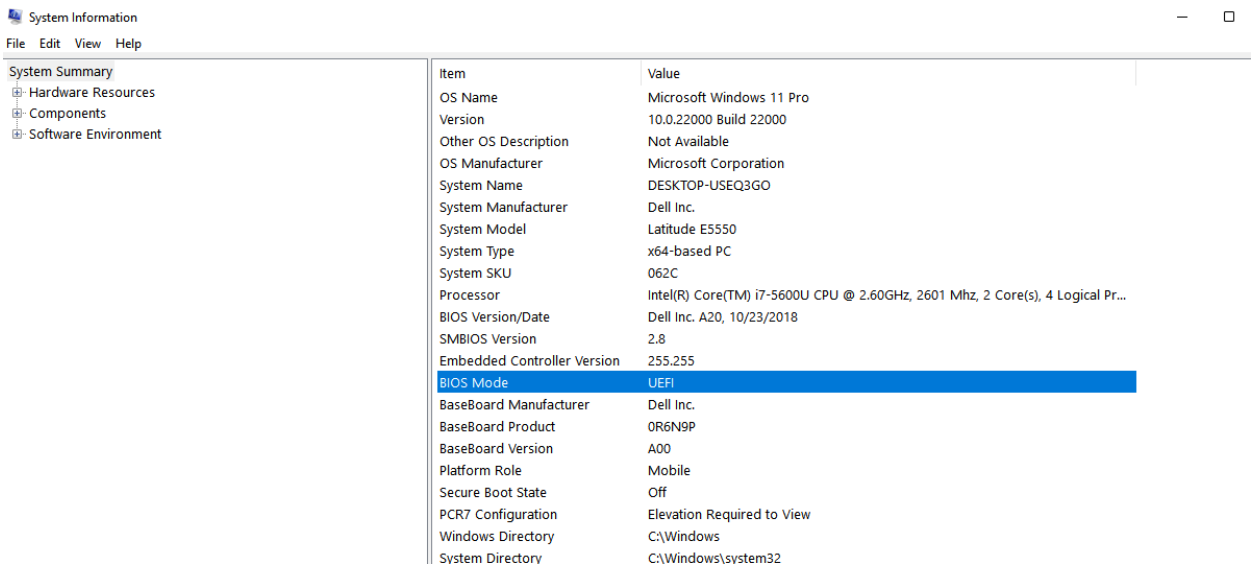
Your explanation must include power-on, POST, BIOS/UEFI, CMOS/firmware settings, MBR/GPT or EFI boot records, boot loader/manager, kernel loading, and operating system control.

State whether your own system uses BIOS/Legacy or UEFI based on system information evidence.

Stage	What Happens	Forensic Relevance
Power-on and POST	Hardware is initialized and checked.	Hardware failure or tampering may affect evidence access.
BIOS/UEFI firmware	Boot device order and firmware settings are applied.	Boot mode can explain partition style and boot artifacts.
MBR/GPT or EFI boot	System locates bootable partition or EFI boot entry.	Boot records may contain evidence of boot configuration.
Boot loader/manager	OS selection and pre-kernel loading occurs.	May show multi-boot or hidden OS behavior.
Kernel and OS control	Operating system fully starts.	Logs and user activity begin to accumulate.

E2. Compare MBR and GPT

Feature	MBR	GPT
Age / design	Older partitioning system from the 1980s.	Modern partitioning system used with UEFI systems.
Size limitation	Commonly limited to about 2 TB.	Supports very large disks; slide notes up to about 9.4 ZB.
Partition information	Partition table is stored in the first sector with boot code.	Uses primary and backup GPT headers and partition entries.
Forensic issue	Critical information in sector 0; damage can affect partition interpretation.	Backup structures may help recovery and validation.



Based on the image above, my system uses UEFI

Part F - File System Identification and Interpretation

For Linux systems, identify whether partitions use ext4, XFS, btrfs, swap, or another file system.

Windows:
wmic logicaldisk get caption,filesystem,size,freespace,volumename
Get-Volume

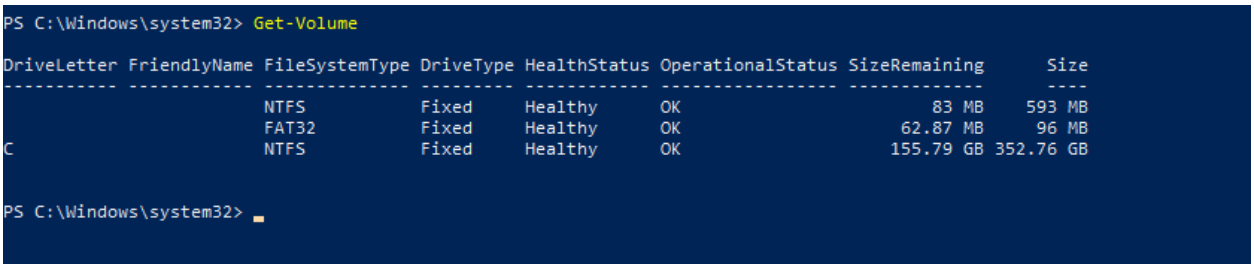
Linux:
lsblk -f
blkid
df -Th

File System	Common OS / Usage	Forensic Importance
FAT/FAT32	Removable media and older systems	Simple structure; often used on USB drives.
NTFS	Modern Windows systems	Includes metadata such as MFT entries, timestamps, permissions, and journals.

ext family	Linux systems	Uses inode-based metadata and supports forensic timeline analysis.
XFS/JFS/btrfs	Linux/Unix-like systems	Advanced features may affect recovery and metadata interpretation.
HFS+	Older macOS systems	Important in Mac forensic investigations; slide notes date support limitations.

Identify the file system used by at least two volumes or partitions on your system.

For Windows systems, identify whether volumes use NTFS, FAT/FAT32, exFAT, or another file system.



a. NTFS and FAT32
b. My file system is **NTFS**

In your report, explain why a file system is important in digital forensics.

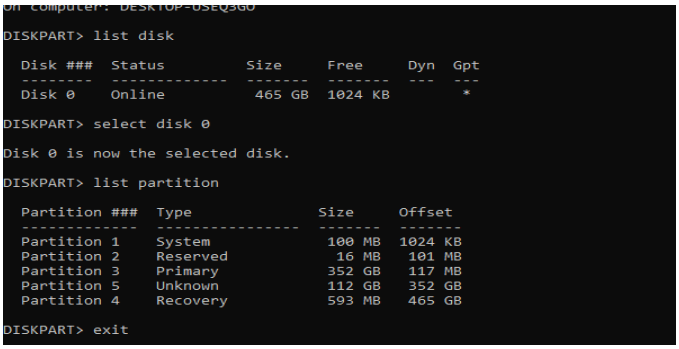
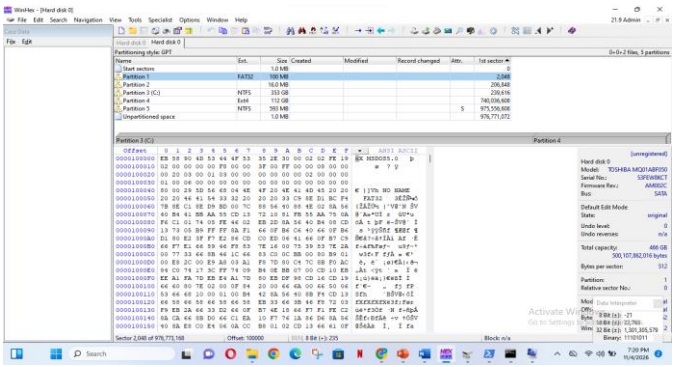
File systems are far more than just a way to organize files and folders. They are a critical source of forensic evidence because they store metadata which is information about the files themselves. This includes creation times, last modification dates, last access timestamps, and owner permissions. When combined, this metadata allows an investigator to build a reliable timeline of user activity. When a file is deleted, the data is not immediately destroyed. In file systems like FAT and NTFS, deletion typically only removes the pointer to the file and marks its storage space as unallocated. The actual data blocks remain intact until they are overwritten. This is what makes the recovery of deleted, lost, or intentionally destroyed files possible. File systems also contain hidden areas that are not visible through the normal operating system

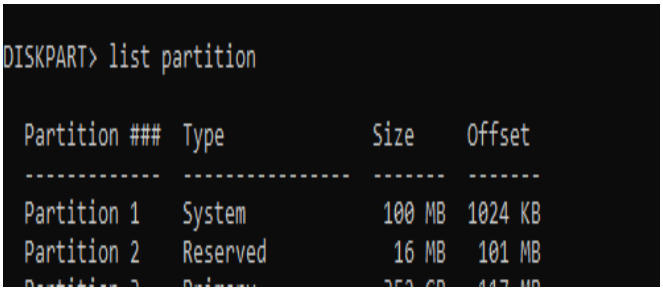
view. These include slack space, unallocated clusters, and in NTFS, Alternate Data Streams (ADS). These structures can be exploited by users or malware to conceal information in plain sight. Additionally, journaling file systems maintain transaction logs that record a history of file operations such as creation, renaming, movement, and deletion. These logs can expose attempts to alter, rename, or destroy evidence, even if the original file is gone. While the operating system presents a clean, user friendly interface of files and folders, a forensic analyst must look beneath it. By understanding the underlying file system architecture, forensic tools can inspect the raw physical medium directly to detect anomalies, tampering, and hidden artifacts that would otherwise go unnoticed.

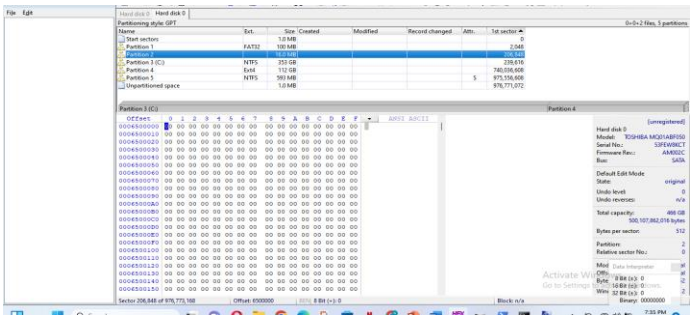
Part G - Final Practical Assignment

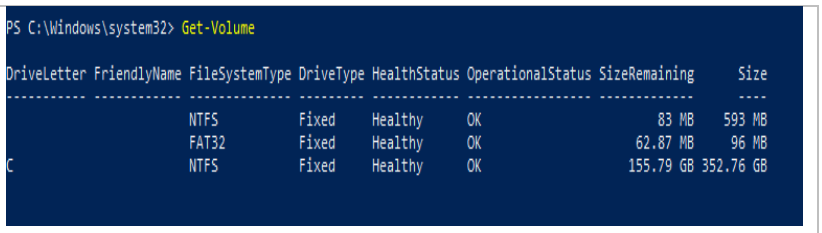
Main assignment

Verify the size of partitions 1 and 2 of a PC using a partition table. Record the verification process step-by-step. Calculate the size of each partition in GB. No full credit will be awarded if you do not use partition table evidence.

Task	Required Evidence	Student Notes / Answer
Identify partition 1	Screenshot of partition table and command output	 

<table>
<tr><td>Record partition 1 sector count</td><td>Start sector, end sector or sector count</td><td>
<table>
<tr><td>Partition identification</td><td>Partition 1</td></tr>
<tr><td>Starting sector</td><td>2,048</td></tr>
<tr><td>Ending sector</td><td>206,847</td></tr>
<tr><td>Sector size</td><td>512 bytes</td></tr>
<tr><td>Sector count</td><td>204,800</td></tr>
</table>
</td></tr>
<tr><td>Calculate partition 1 size</td><td>Formula and result in GB</td><td>Partition size in GB = sector count × 512 ÷ 1,024 ÷ 1,024 ÷ 1,024
Number of sectors = 204,800 sectors.
Size in GB = 204,800 × 512 ÷ 1,024 ÷ 1,024 ÷ 1,024. Partition 1 = 0.0977 GB.</td></tr>
<tr><td>Identify partition 2</td><td>Screenshot of partition table and command output</td><td> 

 </td></tr>
<tr><td>Record partition 2 sector count</td><td>Start sector, end sector or sector count</td><td>
<table>
<tr><th>Required item</th><th>Your findings</th></tr>
<tr><td>Partition identification</td><td>Partition 2</td></tr>
<tr><td>Starting sector</td><td>206,848</td></tr>
<tr><td>Ending sector</td><td>239,615</td></tr>
<tr><td>Sector size</td><td>512 bytes</td></tr>
</table>
</td></tr>
<tr><td>Calculate partition 2 size</td><td>Formula and result in GB</td><td>Partition size in GB = sector_count × 512 ÷ 1,024 ÷ 1,024 ÷ 1,024
Number of sectors = 239,615 - 206,848 + 1 = 32,768 sectors.
Size in GB = 32,768 × 512 ÷ 1,024 ÷ 1,024 ÷ 1,024. Partition 2 = 0.0156 GB.</td></tr>
</table>

Compare with OS-reported size	Disk Management, Get-Volume, lsblk, or df output	
Explain difference if any	Short explanation of rounding, reserved space, file system overhead, or GiB/GB differences	Partition sizes from WinHex were verified against Windows PowerShell (Get-Partition) and both matched.Partition 1 (System): 104,857,600 bytes = ~0.0977 GiBPartition 2 (Reserved): 16,777,216 bytes = ~0.0156 GiBThe agreement between both tools independently confirms the partition sizes are accurate. Any small differences in displayed capacity are due to GB (decimal) vs GiB (binary) conversion.